Section 1: DC Motor Speed Testing

Equipment Used:

- Keysight E3631A DC Power Supply
- Optical tachometer
- DC motors (from mouse kit)

What I Did:
From the "Getting started" section in the project guide, I first calculated how fast the motor needs to turn to complete the 20m track in 10 seconds. The wheel diameter is about 40mm. I then connected one motor to the 6V output of the power supply. I increased the voltage slowly and used the optical tachometer to measure the wheel speed.

How I Used It Correctly:
I set the power supply to the 6V channel and increased the voltage step by step. I held the tachometer close to the wheel without touching it to get a correct reading. The photo of the power supply shows my username (yamae20) on the screen, which proves I used it myself.

Why This Helped Solve the Problem:
This test told us what voltage we need to give the motor to reach a good speed. Without this, we would not know how to design the motor driver circuit.

Section 2: Pick-Up Coil Signal Testing

Equipment Used:

- Tektronix Oscilloscope
- 20kHz track current source
- Pick-up coil (from mouse kit)
- Power supply (for the current source)

What I Did:
As described in the "Pick-up-coil interface design" section, I placed the pick-up coil near the track wire that carries the 20kHz signal. I used the oscilloscope to look at the voltage signal from the coil. I tested with the coil pointing up and sideways to see how the signal changes.

How I Used It Correctly:
I set the oscilloscope to the right voltage range and time scale so I could see the 20kHz wave clearly. I triggered the oscilloscope properly to get a stable picture on the screen.

Why This Helped Solve the Problem:
Seeing the raw sensor signal helped me understand how strong it is and how much noise is there. This told me what kind of filter I needed to design.

Section 3: Bandpass Filter Design and Testing

Equipment Used:

- Tektronix Oscilloscope
- Function generator
- Breadboard and electronic parts (resistors, capacitors)
- Keysight 34401A Multimeter

What I Did:
I designed a bandpass filter to remove noise from the coil signal. I built it on a breadboard. I used the function generator to send a 20kHz signal into the filter (to pretend it was the coil). I used the oscilloscope to check the input and output signals.

How I Used It Correctly:
I set the function generator to 20kHz sine wave with the right voltage. I checked the generator output on the oscilloscope first to make sure it was correct. I used the multimeter to check the resistor and capacitor values before putting them in the circuit.

Why This Helped Solve the Problem:
The filter cleaned up the sensor signal. A clean signal is needed so the mouse can stay on the track. Testing with the function generator proved the filter worked before we used it with the real track.

Section 4: Measuring How Signal Changes with Distance

Equipment Used:

- Tektronix Oscilloscope
- 20kHz track current source
- Pick-up coil with my filter circuit
- Measuring tape

What I Did:

With the filter connected, I moved the coil away from the track at different distances. I used the measuring tape to measure each distance. I recorded the voltage on the oscilloscope at each point.

How I Used It Correctly:

I made sure the coil was at the right height each time. I recorded the voltage only when it was stable on the screen. I wrote down all measurements in a table.

Why This Helped Solve the Problem:

This told us how the signal changes when the mouse moves away from the track centre. This helped us decide where to put the sensor on the mouse and how to write the control code.

Section 5: Testing the Motor Driver Circuit

Equipment Used:

- Keysight E3631A Power Supply
- Tektronix Oscilloscope
- Function generator
- Breadboard with my motor driver circuit
- DC motor

What I Did:

I built the motor driver circuit on a breadboard. I used the function generator to send a control signal to the circuit. I used the oscilloscope to check the output going to the motor. The power supply gave power to the motor.

How I Used It Correctly:

I checked the function generator output on the oscilloscope before connecting it to the circuit. I set the power supply to the voltage we found in Section 1. I used the oscilloscope probes correctly to measure voltage without shorting anything.

Why This Helped Solve the Problem:

The motor driver must give the motor enough power, especially going up hills. Testing on the bench meant we knew it worked before putting it on the mouse.

Section 6: Testing the Full Mouse on the Small Track

Equipment Used:

- Small test track with 20kHz driver
- Oscilloscope
- Multimeter
- Complete mouse (all parts together)

What I Did:

After all parts were tested separately, we put the whole mouse together and tested it on the small practice track. I used the oscilloscope to watch the sensor signal while the mouse moved. I used the multimeter to check the battery voltage.

How I Used It Correctly:
I connected the oscilloscope probes carefully so they did not get caught in the wheels. I watched the signal on the screen to see if noise appeared when the motors were running.

Why This Helped Solve the Problem:
Testing the whole system showed us problems that did not show up when testing parts alone. For example, we saw motor noise getting into the sensor. We fixed this before race day.

Conclusion

I used all the laboratory equipment listed above to help solve the problem of designing a fast line-following mouse. I used each piece of equipment correctly to collect data, test my designs, and find and fix problems. The photos show my username (yamae20) on the equipment, proving I did this work myself. Without using this equipment, I would not have been able to build a mouse that worked well on race day.